

Презентация

Лабораторная работа 6

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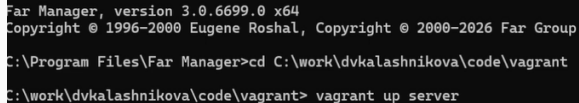
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Информация

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Освоить установку и настройку системы управления базами данных на примере MariaDB.

Поднимаем сервер через vagrant.



```
Far Manager, version 3.0.6699.0 x64  
Copyright © 1996-2000 Eugene Roshal, Copyright © 2000-2026 Far Group  
  
C:\Program Files\Far Manager>cd C:\work\dvkalashnikova\code\vagrant  
C:\work\dvkalashnikova\code\vagrant> vagrant up server
```

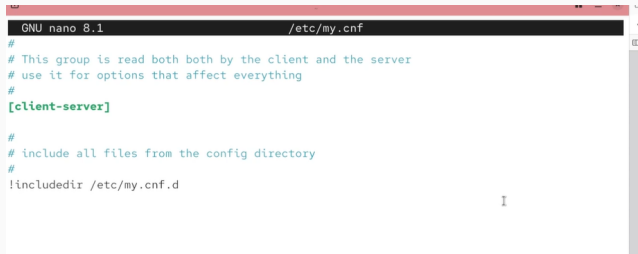
Рис. 1: Запуск сервера

Ставим пакет MariaDB.

```
[dvkalashnikova@server.dvkalashnikova.net ~]$ sudo -i
[sudo] password for dvkalashnikova:
[root@server.dvkalashnikova.net ~]# dnf -y install mariadb mariadb-server
Extra Packages for Enterprise Linux 10 - x86_64      37 kB/s | 40 kB      00:01
Extra Packages for Enterprise Linux 10 - x86_64      2.7 MB/s | 6.7 MB      00:02
Rocky Linux 10 - BaseOS [===                        ] --- B/s | 0 B      --:-- ETA
```

Рис. 2: Установка пакета

Основной конфиг MariaDB: секция client-server и директива !includedir /etc/my.cnf.d.

A screenshot of a terminal window showing the GNU nano 8.1 text editor editing the file /etc/my.cnf. The editor's status bar at the top shows 'GNU nano 8.1' on the left and '/etc/my.cnf' on the right. The file content is as follows:

```
#  
# This group is read both both by the client and the server  
# use it for options that affect everything  
#  
[client-server]  
  
#  
# include all files from the config directory  
#  
!includedir /etc/my.cnf.d
```

The cursor is positioned at the end of the last line. The terminal has a light gray background and a dark gray title bar.

Рис. 3: /etc/my.cnf

Подключение плагина аутентификации GSSAPI (Kerberos); строка закомментирована.



```
GNU nano 8.1 auth_gssapi.cnf
[mariadb]
#plugin-load-add=auth_gssapi.so
```

Рис. 4: auth_gssapi.cnf

Главный конфиг сервера: datadir, socket, log-error, pid-file, секция Galera.

```
GNU nano 0.1 mariadb-server.cnf
#
# These groups are read by MariaDB server.
# Use it for options that only the server (but not clients) should see
#
# See the examples of server my.cnf files in /usr/share/mysql/
#

# this is read by the standalone daemon and embedded servers
[server]

# this is only for the mysql standalone daemon
# Settings user and group are ignored when systemd is used.
# If you need to run mysql under a different user or group,
# customize your systemd unit file for mysql/mariadb according to the
# instructions in http://fedoraproject.org/wiki/Systemd

[mysqld]
datadir=/var/lib/mysql
socket=/var/lib/mysql/mysql.sock
log-error=/var/log/mariadb/mariadb.log
pid-file=/run/mariadb/mariadb.pid

#
#
# * Galera-related settings
#
[galera]
# Mandatory settings
wsrep_on=ON
wsrep_provider=
wsrep_cluster_address=
binlog_format=row
default_storage_engine=InnoDB
innodb_autoinc_lock_mode=2
#
# Allow server to accept connections on all interfaces.
#bind-address=0.0.0.0

Help      Write Out  Where Is  Cut       Execute   Location
Exit      Read File  Replace   Paste     Justify   Go To Line
```

Рис. 5: mariadb-server.cnf

Подключение плагина сжатия LZ4 с принудительной загрузкой.



```
GNU nano 8.1 provider_lz4.cnf
[server]
plugin_load_add=provider_lz4
provider_lz4=force_plus_permanent
```

Рис. 6: provider_lz4.cnf

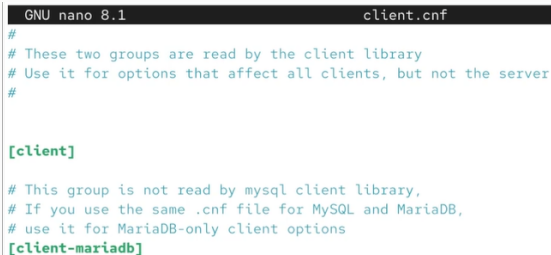
Подключение движка хранения Spider для шардинга; строка закомментирована.



```
GNU nano 8.1 spider.cnf
[mariadb]
#
# Uncomment line to enable
#
#plugin-load-add = ha_spider
# Read more at https://mariadb.com/kb/en/spider/
```

Рис. 7: spider.cnf

Конфиг клиентских приложений: секции client и client-mariadb.

A screenshot of a text editor window titled 'GNU nano 8.1' and 'client.cnf'. The editor shows the following configuration content:

```
#  
# These two groups are read by the client library  
# Use it for options that affect all clients, but not the server  
#  
  
[client]  
  
# This group is not read by mysql client library,  
# If you use the same .cnf file for MySQL and MariaDB,  
# use it for MariaDB-only client options  
[client-mariadb]
```

Рис. 8: client.cnf

Шаблон с пустыми секциями для отдельных утилит MariaDB.

```
GNU nano 8.1                               mysql-clients.cnf
#
# These groups are read by MariaDB command-line tools
# Use it for options that affect only one utility
#

[mysql]

[mysql_upgrade]

[mysqladmin]

[mysqlbinlog]

[mysqlcheck]

[mysqldump]

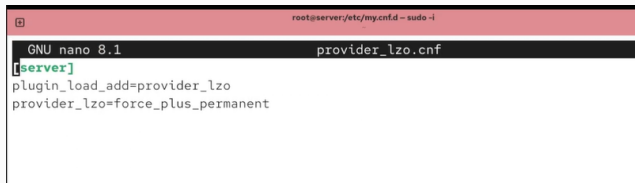
[mysqlimport]

[mysqlshow]

[mysqlslap]
```

Рис. 9: mysql-clients.cnf

Подключение плагина сжатия LZO.



```
root@server:/etc/my.cnf.d - sudo -i
GNU nano 8.1 provider_lzo.cnf
[server]
plugin_load_add=provider_lzo
provider_lzo=force_plus_permanent
```

Рис. 10: provider_lzo.cnf

Пресет для включения шифрования данных Aria, InnoDB, binlog и временных файлов.



```
GNU nano 8.1 enable_encryption.preset
#
# !include this file into your my.cnf (or any of *.cnf files in /etc/my.cnf.d)
# and it will enable data at rest encryption. This is a simple way to
# ensure that everything that can be encrypted will be and your
# data will not leak unencrypted.
#
# DO NOT EDIT THIS FILE! On MariaDB upgrades it might be replaced with a
# newer version and your edits will be lost. Instead, add your edits
# to the .cnf file after the !include directive.
#
# NOTE that you also need to install an encryption plugin for the encryption
# to work. See https://mariadb.com/kb/en/mariadb/data-at-rest-encryption/#encryption-key->
#
[mariadb]
aria-encrypt-tables
encrypt-binlog
encrypt-tmp-disk-tables
encrypt-tmp-files
loose-innodb-encrypt-log
loose-innodb-encrypt-tables
```

Рис. 11: enable_encryption.preset

Подключение плагина сжатия bzip2.



```
GNU nano 8.1 provider_bzip2.cnf
[server]
plugin_load_add=provider_bzip2
provider_bzip2=force_plus_permanent
```

Рис. 12: provider_bzip2.cnf

Подключение плагина сжатия Snappy.



```
GNU nano 8.1 provider_snappy.cnf
[server]
plugin_load_add=provider_snappy
provider_snappy=force_plus_permanent
```

Рис. 13: provider_snappy.cnf

Запускаем службу, ставим в автозагрузку, проверяем порт 3306 через ss.

```
[root@server.dvkalashnikova.net my.cnf.d]# systemctl start mariadb
[root@server.dvkalashnikova.net my.cnf.d]# systemctl enable mariadb
Created symlink '/etc/systemd/system/mysql.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/mysqld.service' → '/usr/lib/systemd/system/mariadb.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/mariadb.service' → '/usr/lib/systemd/system/mariadb.service'.
[root@server.dvkalashnikova.net my.cnf.d]# ss -tulpen | grep mysql
[root@server.dvkalashnikova.net my.cnf.d]# ss -tulpen | grep maria
tcp LISTEN 0      80          0.0.0.0:3306      0.0.0.0:*    users:(("mariadb",pid=18700,fd=18))
        uid:27 ino:89320 sk:1019 cgroup:/system.slice/mariadb.service <->

tcp LISTEN 0      80          [::]:3306       [::]:*      users:(("mariadb",pid=18700,fd=19))
        uid:27 ino:89321 sk:1027 cgroup:/system.slice/mariadb.service v6only:1 <->

[root@server.dvkalashnikova.net my.cnf.d]# █
```

Рис. 14: Запуск mariadb

Выполняем mysql_secure_installation.

```
[root@server.dvkalashnikova.net my.cnf.d]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] n
... skipping.

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] █
```

Подключаемся к базе и выводим справку.

```
[root@server.dvkalashnikova.net my.cnf.d]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 11
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> \h
```

Рис. 16: Подключение к БД

На этом шаге в базе четыре записи.

```
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database          |
+-----+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+-----+
4 rows in set (0.003 sec)

MariaDB [(none)]> █
```

Рис. 17: Списки БД

Версия 10.11.18-MariaDB, id 12, uptime 8 min 2 sec, questions 23, avg 0.047.

```
[root@server.dvkalashnikova.net ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 12
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          12
Current database:
Current user:            root@localhost
SSL:                     Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.11.18-MariaDB MariaDB Server
Protocol version:        10
Connection:              Localhost via UNIX socket
Server characterset:     latin1
Db characterset:         latin1
Client characterset:     utf8mb3
Conn. characterset:      utf8mb3
UNIX socket:             /var/lib/mysql/mysql.sock
Uptime:                  8 min 2 sec

Threads: 1  Questions: 23  Slow queries: 0  Opens: 20  Open tables: 13  Queries per second
avg: 0.047
-----
MariaDB [(none)]> █
```

Рис. 18: Статус БД

Создаём /etc/my.cnf.d/utf8.cnf для смены кодировки.

```
[root@server.dvkalashnikova.net my.cnf.d]# cd /etc/my.cnf.d  
[root@server.dvkalashnikova.net my.cnf.d]# touch utf8.cnf  
[root@server.dvkalashnikova.net my.cnf.d]# nano utf8.cnf
```

Рис. 19: Создание файла конфигурации

Прописываем utf8 как кодировку по умолчанию.



```
GNU nano 8.1 utf8.cnf
[client]
default-character-set = utf8
[mysqld]
character-set-server = utf8
```

Рис. 20: Содержание файла utf8

Перезапускаем mariadb — latin1 меняется на utf8.

```
[root@server.dvkalashnikova.net my.cnf.d]# systemctl restart mariadb
[root@server.dvkalashnikova.net my.cnf.d]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using EditLine wrapper

Connection id:          3
Current database:
Current user:            root@localhost
SSL:                     Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.11.18-MariaDB MariaDB Server
Protocol version:        10
Connection:              Localhost via UNIX socket
Server characterset:     utf8mb3
Db characterset:         utf8mb3
Client characterset:     utf8mb3
Conn. characterset:      utf8mb3
UNIX socket:             /var/lib/mysql/mysql.sock
Uptime:                  31 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.129
-----
```

Рис. 21: Успешная смена кодировки

Создаём БД addressbook, таблицу city и добавляем Иванова, Петрова, Сидорова.

```
MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.010 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.002 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.156 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.012 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.008 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.007 sec)

MariaDB [addressbook]> SELECT * FROM city;
+-----+-----+
| name | city |
+-----+-----+
| Иванов | Москва |
| Петров | Сочи |
| Сидоров | Дубна |
+-----+-----+
3 rows in set (0.001 sec)

MariaDB [addressbook]> █
```

Рис. 22: Наполнение таблицы

Создаём пользователя dvkalashnikova и даём доступ к БД.

```
MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO dvkalashnikova@"%";
Query OK, 0 rows affected (0.006 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.004 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+
2 rows in set (0.005 sec)

MariaDB [addressbook]> quit
Bye
[root@server.dvkalashnikova.net my.cnf.d]# mysqlshow -u root -p
Enter password:
+-----+
| Databases |
+-----+
| addressbook |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
[root@server.dvkalashnikova.net my.cnf.d]# mysqlshow -u root -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+
[root@server.dvkalashnikova.net my.cnf.d]#
```

Рис. 23: Проверка прав доступа для нового пользователя

Делаем обычный, сжатый и с таймстемпом бэкапы, сохраняем конфиги.

```
[root@server.dvkalashnikova.net my.cnf.d]# mkdir -p /var/backup
[root@server.dvkalashnikova.net my.cnf.d]# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
[root@server.dvkalashnikova.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
[root@server.dvkalashnikova.net my.cnf.d]# mysqldump -u root -p addressbook | gzip > $(date +%Y%m%d.%H%M%S).sql.gz)
Enter password:
[root@server.dvkalashnikova.net my.cnf.d]# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
[root@server.dvkalashnikova.net my.cnf.d]# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
[root@server.dvkalashnikova.net my.cnf.d]# cd /vagrant/provision/server
[root@server.dvkalashnikova.net server]# mkdir -p /vagrant/provision/server/mysql/etc/my.cnf.d
[root@server.dvkalashnikova.net server]# mkdir -p /vagrant/provision/server/mysql/var/backup
[root@server.dvkalashnikova.net server]# cp -R /etc/my.cnf.d/utf8.cnf /vagrant/provision/server/mysql/etc/my.cnf.d/
[root@server.dvkalashnikova.net server]# cp -R /var/backup/* /vagrant/provision/server/mysql/var/backup/
[root@server.dvkalashnikova.net server]# touch mysql.sh
[root@server.dvkalashnikova.net server]# chmod +x mysql.sh
[root@server.dvkalashnikova.net server]# nano mysql.sh
[root@server.dvkalashnikova.net server]# █
```

Рис. 24: Бэкапы и сохранение vagrant

Прописываем в скрипте команды настройки БД.

```
GNU nano 8.1                                mysql.sh                                Modif
#!/bin/bash
:cho "Provisioning script $0"
systemctl restart named
:cho "Install needed packages"
inf -y install mariadb mariadb-server
:cho "Copy configuration files"
:p -R /vagrant/provision/server/mysql/etc/* /etc
:mkdir -p /var/backup
:p -R /vagrant/provision/server/mysql/var/backup/* /var/backup
:cho "Start mysql service"
systemctl enable mariadb
systemctl start mariadb
if [[ ! -d /var/lib/mysql/mysql ]]
:then
:cho "Securing mariadb"
mysql_secure_installation <<EOF
/
123456
123456
/
/
/
/
:EOF
:cho "Create database"
mysql -u root -p123456 <<EOF
CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
:EOF
mysql -u root -p123456 addressbook < /var/backup/addressbook.sql
:q
```

Добавляем автозагрузку mysql.sh.

```
server.vm.provision "server dummy", type: "shell", preserve_order: true, path: "provision/server/01-dummy.sh"
server.vm.provision "server forward", type: "shell", preserve_order: true, run: "always", path: "provision/server/02-forward.sh"
server.vm.provision "server dns", type: "shell", preserve_order: true, path: "provision/server/dns.sh"
server.vm.provision "server dhcp", type: "shell", preserve_order: true, path: "provision/server/dhcp.sh"
server.vm.provision "server http", type: "shell", preserve_order: true, path: "provision/server/http.sh"
server.vm.provision "server mysql", type: "shell", preserve_order: true, path: "provision/server/mysql.sh"
end

## Client configuration
```

Рис. 26: Vagrantfile

Получены навыки работы с базами данных и их настройкой.

1. MariaDB Foundation. MariaDB Documentation [Электронный ресурс]. — URL: <https://mariadb.com/kb/en/documentation/> (дата обращения: 07.10.2026).
2. MariaDB Foundation. MariaDB Server Configuration Files [Электронный ресурс]. — URL: <https://mariadb.com/kb/en/configuring-mariadb-with-option-files/> (дата обращения: 07.10.2026).
3. Red Hat. Documentation [Электронный ресурс]. — URL: <https://docs.redhat.com/> (дата обращения: 07.10.2026).